

Modified Gradient Bandit with Adaptive Baseline

RL Assignment 1 Report

Ashwin K.M.

Abstract

This report studies a modified Gradient Bandit algorithm equipped with an adaptive, variance-aware reward baseline for non-stationary environments. While the standard gradient bandit uses the running average reward as a baseline, this can lag behind when reward distributions drift over time. To address this, we introduce an adaptive baseline that combines the global average reward with a recent variance-adjusted mean, controlled by a mixing parameter β . Experiments are conducted on a non-stationary 10-armed bandit where arm means undergo random-walk drift every 500 steps. Empirically, adaptive baselines ($\beta = 0.3, 0.6$) improve reward and reduce regret compared to the standard baseline ($\beta = 0$). Notably, higher β values enable faster adaptation under drift, highlighting the benefit of variance-sensitive baseline tracking.

1 Problem Statement

We consider the Gradient Bandit problem in a *non-stationary* reward setting. The gradient bandit uses a baseline (typically the running average reward) to reduce the variance of preference updates. However, in drifting environments where reward distributions change over time, a simple global average may fail to reflect the current reward landscape.

To improve robustness under drift, we study an **adaptive variance-based baseline** that adjusts automatically using recent reward statistics. We evaluate performance across multiple baseline mixing values $\beta \in \{0, 0.3, 0.6\}$, tracking reward, optimal action selection, baseline evolution, and cumulative regret.

2 Methodology

2.1 Gradient Bandit Algorithm

The gradient bandit maintains a real-valued preference $H[a]$ for each action (arm) a . Actions are selected probabilistically according to a softmax policy:

$$\pi[a] = \frac{\exp(H[a])}{\sum_b \exp(H[b])}. \quad (1)$$

After receiving reward R , the preferences are updated via gradient ascent on expected reward.

For the selected action A_t :

$$H[A_t] \leftarrow H[A_t] + \alpha(R - b_t)(1 - \pi[A_t]). \quad (2)$$

For all non-selected actions $a \neq A_t$:

$$H[a] \leftarrow H[a] - \alpha(R - b_t)\pi[a]. \quad (3)$$

Here, α is the step-size and b_t is the baseline used to reduce update variance.

2.2 Adaptive Variance-Based Baseline

A key modification is the adaptive baseline. Instead of relying only on the running average reward, we propose a variance-sensitive baseline of the form:

$$b_t = (1 - \beta) \bar{R}_t + \beta \hat{R}_t^{(var)}. \quad (4)$$

Where:

- $\bar{R}_t$ is the running mean of all rewards,
- $\hat{R}_t^{(var)}$ is a variance-adjusted recent mean over the last W rewards,
- $\beta \in [0, 1]$ controls adaptivity.

The recent variance-adjusted mean weights rewards inversely proportional to squared deviation:

$$w_i = \frac{1}{(r_i - \mu_W)^2 + \epsilon}, \quad \hat{R}_t^{(var)} = \frac{\sum_i w_i r_i}{\sum_i w_i}. \quad (5)$$

Thus, higher β values emphasize recent stable reward trends, allowing faster adaptation under non-stationarity.

2.3 Non-Stationary Bandit Environment

Experiments use a 10-armed bandit whose true means drift over time:

- True means initialized as $\mu_a \sim \mathcal{N}(0, 1)$.
- Every 500 steps, drift is applied:

$$\mu_a \leftarrow \mu_a + \mathcal{N}(0, 0.1).$$

- Rewards are sampled as:

$$R \sim \mathcal{N}(\mu_a, 1).$$

This environment tests whether adaptive baselines can track a changing reward landscape.

3 Experimental Setup

All experiments use the following parameters:

- Number of runs: $n_{runs} = 200$
- Time steps per run: $n_{steps} = 2000$
- Arms: $n_{arms} = 10$
- Step size: $\alpha = 0.1$
- Baseline window: $W = 50$
- Drift interval: every 500 steps

- Drift magnitude: $\sigma_{walk} = 0.1$

We compare:

$$\beta = 0.0 \text{ (standard)}, \quad \beta = 0.3, \quad \beta = 0.6.$$

Performance metrics include average reward, optimal action selection, baseline evolution, and cumulative regret:

$$\text{Regret}_t = \mu^* - \mu_{\text{chosen}}.$$

Reward and optimal-action curves are plotted with a running average window of 100 steps for smoother visualization.

4 Results and Discussion

4.1 Average Reward Over Time

Figure 1 shows that adaptive baselines consistently improve reward compared to the standard gradient bandit ($\beta = 0$). Notably, $\beta = 0.6$ achieves the highest long-term reward, suggesting stronger responsiveness to recent reward changes. The standard baseline learns more slowly because the global average reward lags behind shifting arm means. Both $\beta = 0.3$ and $\beta = 0.6$ adapt faster after distribution changes, indicating better tracking of drifting reward dynamics.

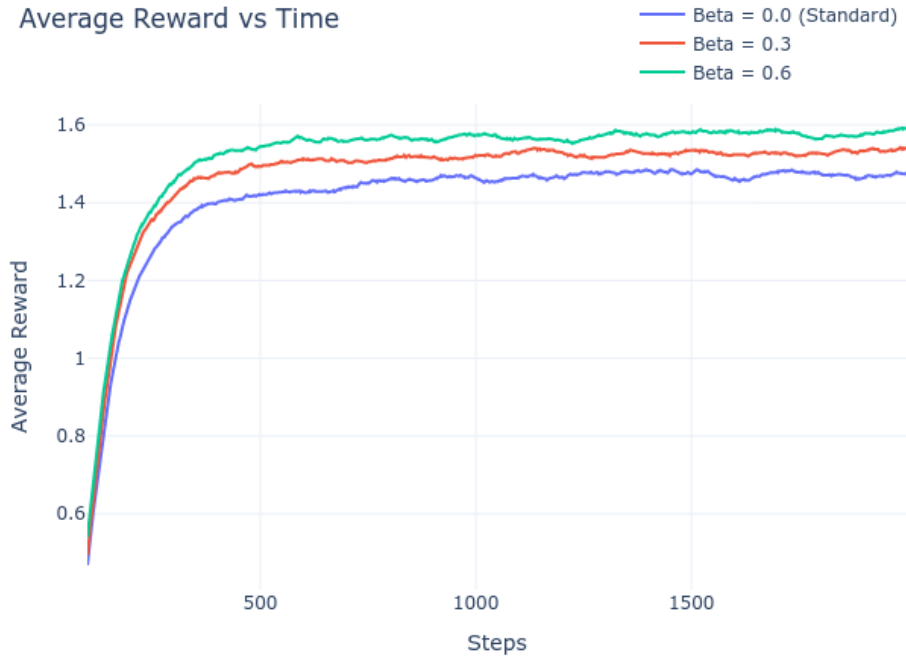


Figure 1: Average reward over time for different baseline mixing values β .

4.2 Optimal Action Selection

Figure 2 reports the percentage of times the agent selects the current optimal arm. The adaptive methods substantially outperform $\beta = 0$, reaching higher optimal-action rates earlier in training. This improvement is especially visible after drift points, where adaptive baselines help re-identify the best arm more quickly. Interestingly, $\beta = 0.3$ slightly edges out $\beta = 0.6$ in stability, indicating that very high adaptivity can introduce mild variance.

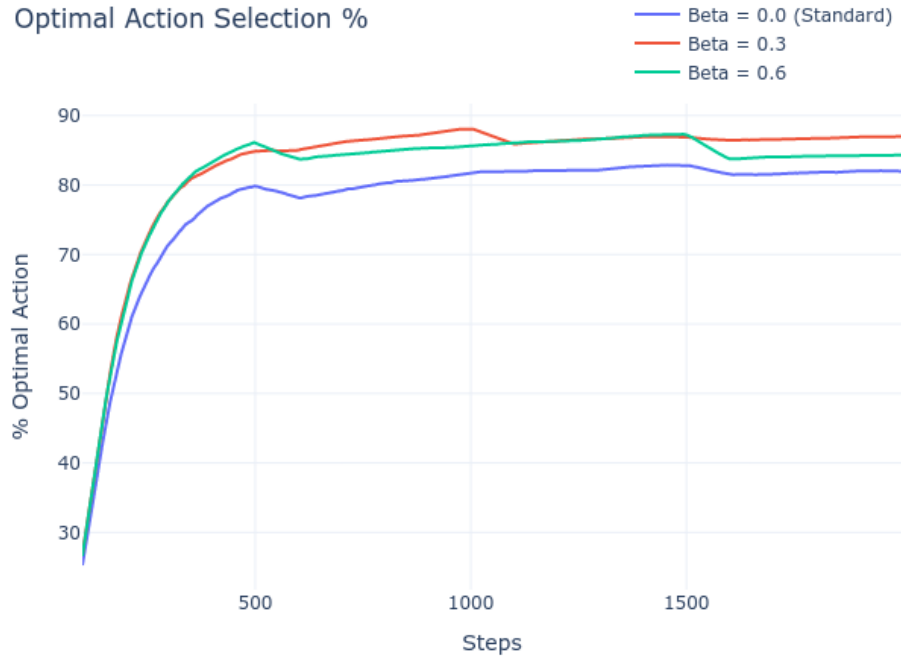


Figure 2: Optimal action selection percentage over time.

4.3 Baseline Evolution

Figure 3 illustrates how the baseline evolves over time. For $\beta = 0$, the baseline increases smoothly as a slow running average, which can under-react to sudden reward shifts. In contrast, $\beta = 0.3$ and $\beta = 0.6$ rise more quickly and track reward changes more aggressively. Notably, $\beta = 0.6$ produces the most responsive baseline, aligning closely with the drifting reward landscape.

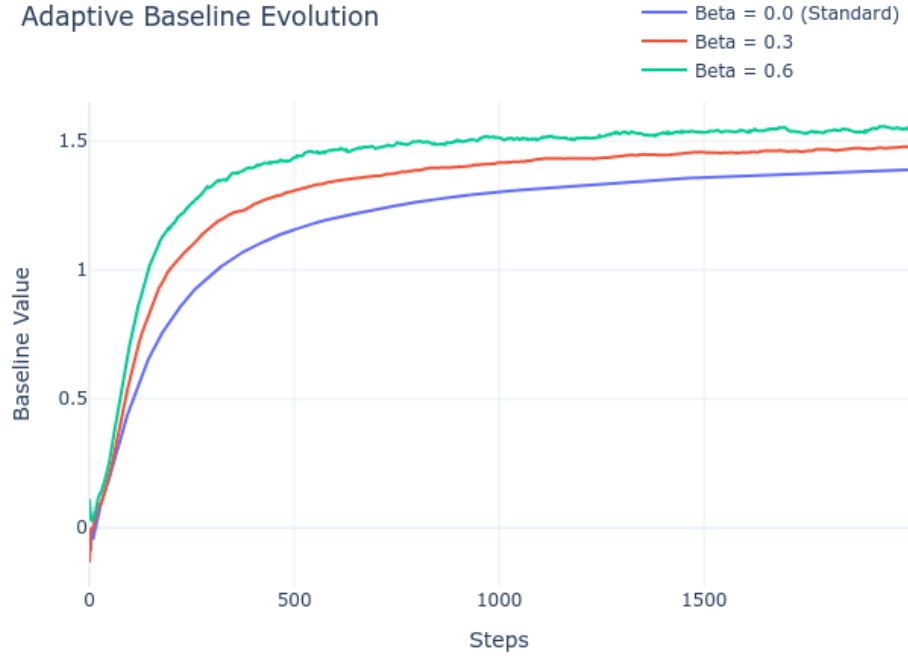


Figure 3: Adaptive baseline evolution across different β values.

4.4 Cumulative Regret

Figure 4 presents cumulative regret, computed as the difference between the optimal expected reward and the chosen action's expected reward. The standard baseline ($\beta = 0$) accumulates regret more rapidly, indicating slower adaptation to drift. Adaptive baselines reduce regret growth, with $\beta = 0.6$ achieving the lowest regret overall. This highlights that variance-aware baselines improve learning efficiency under non-stationary conditions.



Figure 4: Cumulative regret over time under non-stationary drift.

5 Conclusions

These experiments confirm that adaptive variance-based baselines outperform the standard gradient bandit baseline in non-stationary environments. Across reward, optimal-action selection, and regret metrics, $\beta = 0.3$ and $\beta = 0.6$ consistently improve performance. Higher β values yield faster adaptation to drift but may introduce slightly more baseline variance. Overall, the adaptive baseline successfully tracks the changing reward landscape, reducing regret and improving learning robustness.

6 Assumptions and Implementation Notes

The implementation follows the described methodology with the following assumptions:

- Softmax probabilities are stabilized numerically using $\max(H)$ subtraction.
- $\beta = 0$ reduces the baseline exactly to the running average reward.
- Variance weighting uses $\epsilon = 10^{-8}$ for numerical stability.
- Drift is applied exactly every 500 steps via additive $\mathcal{N}(0, 0.1)$ noise.
- Reward noise is fixed as $\mathcal{N}(\mu_a, 1)$.
- Reward and optimal-action plots use a moving average window of 100 steps.
- Regret is computed using the difference in true means between the optimal and selected actions.

Symbol	Description
Bandit Actions & Policy	
$a \in \{1, \dots, n_{\text{arms}}\}$	Action (arm) index
A_t	Action selected at time step t
$H[a]$	Preference value associated with action a
$\pi[a]$	Softmax probability of selecting action a
α	Step size (learning rate) for preference updates
Rewards & Baseline Terms	
R	Reward observed after selecting an action
$\bar{R}_t$	Running average reward up to time t
b_t	Baseline used in gradient bandit update
$\beta \in [0, 1]$	Mixing parameter controlling baseline adaptivity
W	Window size for recent reward statistics
r_i	Reward sample within the last W steps
μ_W	Mean of the most recent W rewards
$\hat{R}_t^{(\text{var})}$	Variance-adjusted recent mean reward
w_i	Weight inversely proportional to squared deviation of r_i
ε	Small constant for numerical stability (10^{-8})
Non-Stationary Environment	
μ_a	True (unknown) expected reward of arm a
σ_{walk}	Standard deviation of random-walk drift in arm means
$\mathcal{N}(\mu, 1)$	Reward distribution with mean μ and unit variance
Regret & Optimality	
μ^*	Expected reward of the optimal arm at time t
μ_{chosen}	Expected reward of the selected arm
Regret_t	Instantaneous regret: $\mu^* - \mu_{\text{chosen}}$
Experiment Hyperparameters	
n_{arms}	Number of arms (actions) in the bandit
n_{runs}	Number of independent experimental runs
n_{steps}	Number of time steps per run

Table 1: Notation reference for mathematical symbols and hyperparameters used in this report.